

Lie Group Algebra: $SE(3)$

Transformation & Rotations

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1 Introduction

This document covers the theory behind Lie Groups, Rotations and Transformations. This serves as an introductory course to the maths and shows examples of workflows and implementation.

2 Coordinate Frames

Suppose we have 2 coordinate frames, frame A and frame B. Let's say we can describe the rotation from frame A to frame B through a rotation matrix, ${}^B\mathbf{R}_A \in \mathbb{R}^{3 \times 3}$ and a translation vector ${}^B\mathbf{t}_A \in \mathbb{R}^3$. And let's say we have a point in space described in frame A, ${}^A\mathbf{p}$. We can then translate the point into frame B, ${}^B\mathbf{p}$, via:

$${}^B\mathbf{p} = {}^B\mathbf{R}_A {}^A\mathbf{p} + {}^B\mathbf{t}_A \quad (2.1)$$

First thing to notice is that a rigid body transformation is both a rotation and a translation. The definition of the variables in 2.1 are as follows:

- ${}^B\mathbf{p}$: Point expressed in frame B.
- ${}^B\mathbf{R}_A$: Rotation matrix from frame A to frame B.
- ${}^A\mathbf{p}$: Point expressed in frame A.
- ${}^B\mathbf{t}_A$: Translation from frame A to frame B **expressed** in frame B.

It is important to get a clear understanding of what the variables mean and what frames they are in. Figure 1 shows a diagram of the different vectors.

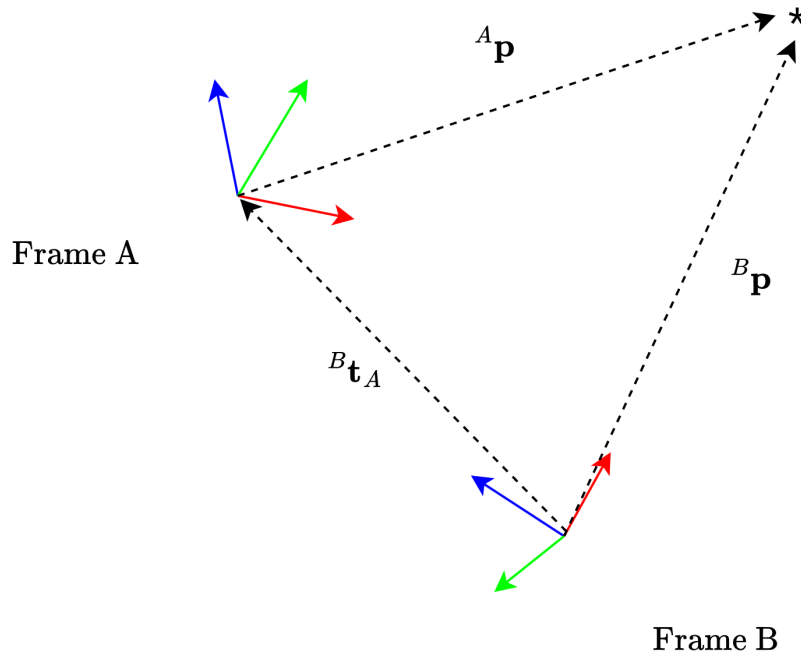


Figure 1: Figure showing the transformation of points between two coordinate frames A and B.

3 SO(3): Rotation

A 3D point can be expressed by a 3×3 rotation matrix:

$$\mathbf{R} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \quad (3.1)$$

A valid rotation matrix must satisfy the following constraints:

$$\mathbf{R}\mathbf{R}^T = \mathbf{R}^T\mathbf{R} = \mathbb{I}_{3 \times 3} \quad (3.2)$$

$$\det(\mathbf{R}) = 1 \quad (3.3)$$

All matrices which satisfy the constraints 3.2 and 3.3 are denoted as **Special Orthogonal** (SO) matrices. With $SO(3)$ denoting a 3 dimensional rotational space. This can be written as:

$$SO(3) = \{\mathbf{R} \in \mathbb{R}^{3 \times 3} : \mathbf{R}\mathbf{R}^T = \mathbb{I}_{3 \times 3}, \det(\mathbf{R}) = 1\} \quad (3.4)$$

3.1 Why is SO(3) a group?

The special orthogonal group $SO(3)$ is a group because the composition of rotation matrices satisfies the four group properties: closure, identity, inverse, and associativity.

Composition

The composition of two rotation matrices produces another valid rotation matrix. Therefore, for $\mathbf{R}_1, \mathbf{R}_2 \in SO(3)$:

$$\mathbf{R}_3 = \mathbf{R}_2\mathbf{R}_1 \in SO(3) \quad (3.5)$$

Identity

There exists an identity rotation that leaves every vector unchanged. This is represented by the identity matrix $\mathbb{I}_{3 \times 3}$:

$$\mathbf{R}\mathbf{I} = \mathbf{I}\mathbf{R} = \mathbf{R} \quad (3.6)$$

Inverse

Every rotation can be undone by applying its inverse. For rotation matrices, the inverse is equal to the transpose:

$$\mathbf{R}^{-1} = \mathbf{R}^T \quad (3.7)$$

Associativity

The grouping of multiple rotations does not affect the resulting rotation:

$$(\mathbf{R}_1\mathbf{R}_2)\mathbf{R}_3 = \mathbf{R}_1(\mathbf{R}_2\mathbf{R}_3) \quad (3.8)$$

Therefore, $SO(3)$ forms a group. The important additional property is that $SO(3)$ is also a smooth mathematical manifold. This is what makes $SO(3)$ a Lie group. A general 3×3 matrix has 9 independent elements. However, a rotation matrix is constrained by the orthogonality and determinant conditions defined above. These constraints reduce the number of independent degrees of freedom to 3. This structure allows rotations to be represented locally using a corresponding Lie algebra.

3.1.1 Axis-Angle Representation

Every 3D rotation can be described with:

- An axis of rotation $\vec{u}$ where, $|\vec{u}| = 1$.
- An angle of rotation θ , measured in radians.

We can combine these into a single vector called the rotation vector $\vec{\phi}$:

$$\vec{\phi} = \theta \vec{u} \quad (3.9)$$

which has the properties:

- $|\vec{\phi}| = \theta$
- $\frac{\vec{\phi}}{\theta} = \vec{u}$

Hence, a 3D rotation can be described by a 3D vector.

3.1.2 The Hat Operator

Given:

$$\vec{\phi} = \begin{bmatrix} \phi_x \\ \phi_y \\ \phi_z \end{bmatrix} \quad (3.10)$$

we can define:

$$\wedge : \mathbb{R}^3 \rightarrow \mathcal{SO}(3) \quad (3.11)$$

as:

$$\vec{\phi}^\wedge = \begin{bmatrix} 0 & -\phi_z & \phi_y \\ \phi_z & 0 & -\phi_x \\ -\phi_y & \phi_x & 0 \end{bmatrix} \quad (3.12)$$

This matrix has the important property:

$$\vec{\phi}^\wedge \vec{x} = \vec{\phi} \times \vec{x} \quad (3.13)$$

The set of all matrices of the form $\vec{\phi}^\wedge$ is called $\mathcal{SO}(3)$.

$\mathcal{SO}(3)$: Contains actual rotation

$\mathcal{SO}(3)$: Contains infinitesimal rotations / rotation vector represented as matrices.

The exponential and logarithmic map connect the two together.

3.2 The Exponential Map

Suppose we have a rotation vector, $\vec{\phi}$. We can use the exponential map to transform it into a rotation matrix:

$$\mathbf{R} = \exp(\vec{\phi}^\wedge) \quad (3.14)$$

This is the Lie group exponential map:

$$\exp : \mathcal{SO}(3) \rightarrow \text{SO}(3) \quad (3.15)$$

The exponential map can be defined using the matrix exponential:

$$\exp(\vec{\phi}^\wedge) = \mathbf{I}_{3 \times 3} + \vec{\phi}^\wedge + \frac{1}{2!} (\vec{\phi}^\wedge)^2 + \frac{1}{3!} (\vec{\phi}^\wedge)^3 + \dots \quad (3.16)$$

This infinite series has a closed-form solution called the Rodrigues rotation formula.

$$\theta = \|\vec{\phi}\| \quad (3.17)$$

$$\mathbf{R} = \mathbb{I}_{3 \times 3} + \frac{\sin \theta}{\theta} \vec{\phi}^\wedge + \frac{1 - \cos \theta}{\theta^2} (\vec{\phi}^\wedge)^2 \quad (3.18)$$

This converts:

$$\vec{\phi} \rightarrow \vec{\phi}^\wedge \rightarrow \mathbf{R} \quad (3.19)$$

3.3 The Logarithmic Map

We can go the other way as well. Suppose we have a rotation matrix, $\mathbf{R}$. We can use the logarithmic map to transform it into a rotation vector:

$$\vec{\phi} = \log(\mathbf{R}) \quad (3.20)$$

This is the Lie group logarithmic map:

$$\log : \text{SO}(3) \rightarrow \mathcal{SO}(3) \quad (3.21)$$

The logarithmic map can be defined using the following:

$$\theta = \cos^{-1} \left(\frac{\text{tr}(\mathbf{R}) - 1}{2} \right) \quad (3.22)$$

$$\vec{\phi} = \frac{\theta}{2 \sin \theta} (\mathbf{R} - \mathbf{R}^T) \quad (3.23)$$

4 SE(3): Translation

4.1 SE(3) group

Now we can add translation:

$$\mathbf{T} = \begin{bmatrix} \mathbf{R} & \vec{t} \\ 0 & 1 \end{bmatrix} \quad (4.1)$$

Where:

$$\mathbf{R} \in \mathbb{R}^{3 \times 3} \quad (4.2)$$

$$\vec{t} \in \mathbb{R}^3 \quad (4.3)$$

The set of all transformation is SE(3). Formally it is:

$$\text{SE}(3) = \left\{ \begin{bmatrix} \mathbf{R} & \vec{t} \\ 0 & 1 \end{bmatrix} : \mathbf{R} \in \text{SO}(3), \vec{t} \in \mathbb{R}^3 \right\} \quad (4.4)$$

This is the group of 3D rigid-body transformations.

4.2 Homogeneous Coordinates

Normally:

$${}^B\vec{p} = {}^B\mathbf{R}_A {}^A\vec{p} + {}^B\vec{t}_A \quad (4.5)$$

The rotation and translation are separate. Using homogeneous coordinates:

$$\tilde{p} = \begin{bmatrix} \vec{p} \\ 1 \end{bmatrix} \quad (4.6)$$

Then we can write:

$${}^B\tilde{p} = {}^B\mathbf{T}_A {}^A\tilde{p} \quad (4.7)$$

This then allows:

$${}^C\mathbf{T}_A = {}^C\mathbf{T}_B {}^B\mathbf{T}_A \quad (4.8)$$

4.3 SE(3) Inverse

If:

$${}^B\mathbf{T}_A = \begin{bmatrix} {}^B\mathbf{R}_A & {}^B\vec{t}_A \\ 0 & 1 \end{bmatrix} \quad (4.9)$$

Then the inverse is:

$${}^B\mathbf{T}_A^{-1} = \begin{bmatrix} {}^B\mathbf{R}_A^T & -{}^B\mathbf{R}_A^T {}^B\vec{t}_A \\ 0 & 1 \end{bmatrix} \quad (4.10)$$

Notice that the translation is not simply $-{}^B\vec{t}_A$. It must first be rotated.

4.4 $\mathcal{SE}(3)$ Lie Algebra

Just as $\text{SO}(3)$ has its Lie algebra, $\mathcal{SO}(3)$, $\text{SE}(3)$ has:

$$\mathcal{SE}(3) \quad (4.11)$$

An element of algebra can be represented by a 6-vector:

$$\xi = \begin{bmatrix} \vec{\rho} \\ \vec{\phi} \end{bmatrix} \quad (4.12)$$

where $\vec{\rho}$ is translation and $\vec{\phi}$ is rotation:

$$\vec{\rho} = \begin{bmatrix} \rho_x \\ \rho_y \\ \rho_z \end{bmatrix} \quad (4.13)$$

$$\vec{\phi} = \begin{bmatrix} \phi_x \\ \phi_y \\ \phi_z \end{bmatrix} \quad (4.14)$$

We can then:

$$\xi^\wedge = \begin{bmatrix} \phi^\wedge & \vec{\rho} \\ 0 & 0 \end{bmatrix} \quad (4.15)$$

which takes:

$$\mathbb{R}^6 \rightarrow \mathcal{SO}(3) \quad (4.16)$$

4.5 SE(3) Exponential Map

The exponential map is defined as:

$$T = \exp(\xi^\wedge) \quad (4.17)$$

The result is:

$$\exp(\xi^\wedge) = \begin{bmatrix} \exp({}^B\vec{\phi}_A) & V^B\vec{\rho}_A \\ 0 & 1 \end{bmatrix} \quad (4.18)$$

where:

$$R = \exp(\phi^\wedge) \quad (4.19)$$

$$V = \mathbb{I} + \frac{1 - \cos(\theta)}{\theta^2} \phi^\wedge + \frac{\theta - \sin(\theta)}{\theta^3} (\phi^\wedge)^2 \quad (4.20)$$

This is slightly more subtle than SO(3), because translation and rotation are coupled in the exponential map.

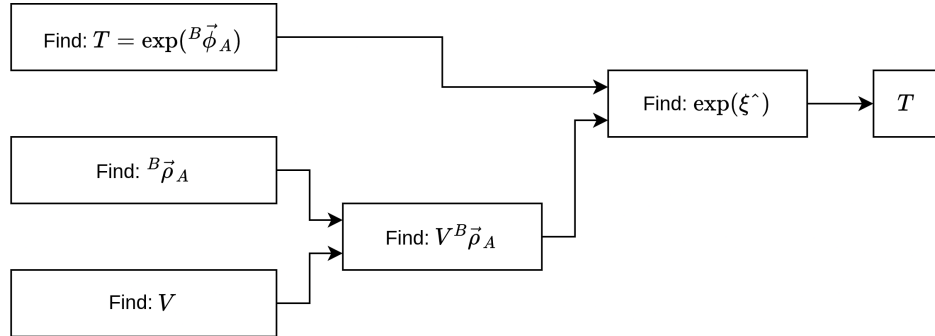


Figure 2: Figure showing the process to find the transform matrix through the exponential map.

4.6 SE(3) Logarithmic Map

Going backwards:

$${}^B T_A = \begin{bmatrix} {}^B R_A & {}^B \vec{t}_A \\ 0 & 1 \end{bmatrix} \quad (4.21)$$

We first calculate:

$${}^B \vec{\phi}_A = \log(R) \quad (4.22)$$

Then:

$${}^B \vec{\rho}_A = V^{-1} {}^B \vec{t}_A \quad (4.23)$$

giving:

$$\xi = \begin{bmatrix} \vec{\rho} \\ \vec{\phi} \end{bmatrix} \quad (4.24)$$

therefore:

$$\log({}^B T_A) = {}^B \xi_A^\wedge \quad (4.25)$$

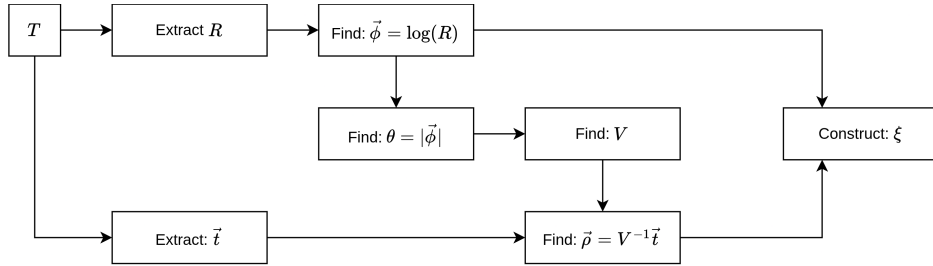
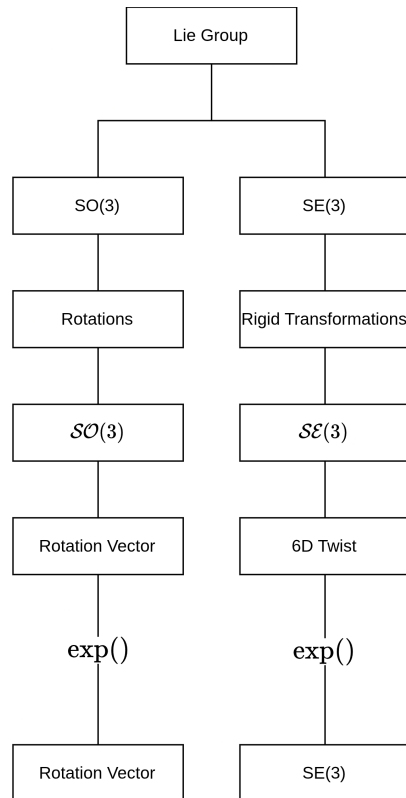


Figure 3: Figure showing the process to find the twist vector through the logarithmic map.

5 The Big Picture



$$\mathcal{SO}(3) - \exp() \rightarrow SO(3)$$

$$\mathcal{SE}(3) - \exp() \rightarrow SE(3)$$

Figure 4: Figure showing the processes for lie group algebra.

5.1 Why robotics use this?

The key advantage is that $SE(3)$ is a constrained space, while its Lie Algebra is essentially an unconstrained vector space. $R \in SO(3)$ cannot be $R + \Delta R$, but we can represent a small update as:

$$\delta \vec{\phi} \in \mathbb{R}^3 \quad (5.1)$$

and update:

$$R_{new} = \exp(\delta \hat{\phi}) R \quad (5.2)$$

The $\exp()$ guarantees that the result remains in $SO(3)$. Similarly:

$$T_{new} = \exp(\delta \hat{\xi}) T \quad (5.3)$$

5.2 C++ Eigen Library

Eigen is widely used in robotics, SLAM, computer vision, and estimation for linear algebra.

```
Eigen::Matrix3d      R;  
Eigen::Matrix4d      T;  
Eigen::Vector3d      phi;  
Eigen::Vector<double, 6> xi;
```

Eigen itself is not JSF compliant. However, you can use it in JSF source code as a third party library. In section 4.5 of JSF standard covers libraries. AV rule 16 "Only DO-178B level A certifiable or SEAL 1 C/C++ libraries shall be used with safety critical code". Hence you can use it, but with care.

6 Active Vs Passive Transformation

Active: Rotate the physical object / vector Passive: Rotate the frame and keep the vector constant
The mathematics is the same but the interpretation is different.

6.1 Active vs Passive in Lie Groups

Active:

$$R_{new} = \exp(\hat{\vec{\phi}}) R \quad (6.1)$$

Passive:

$$R_{new} = R \exp(-\hat{\vec{\phi}}) \quad (6.2)$$

Passive transformation is the inverse of an active transform.

Usually in robotics we describe the 'pose' of a robot using the passive transformation from body coordinates to world coordinates.

$${}^w\vec{p} = {}^wT_b {}^b\vec{p} \quad (6.3)$$

So, if you have points in the robots body frame you can transform them into the world frame. As you can observe things from the robots POV, this saves you having to inverse when converting things to a world frame.